IT UNIVERSITY OF COPENHAGEN

Game Programming 25

L02: Collisions

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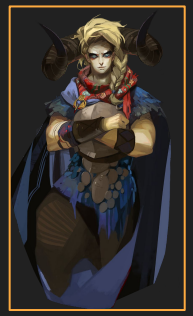
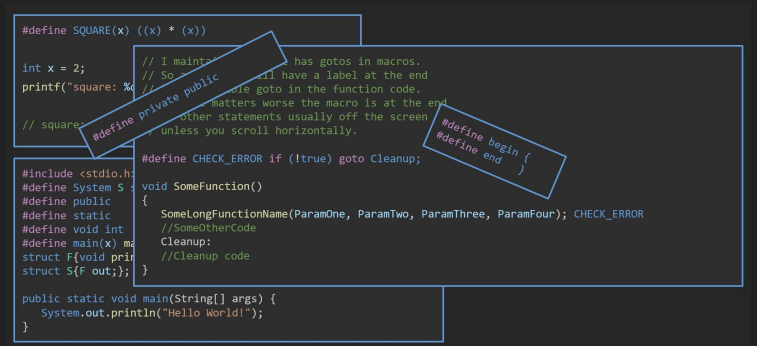
Recap



images and textures

sprites



preprocessor directives and macros

C++ memory layout

Exercise 01 review - Add projectiles (init)

```
// asteroids init
{
    for(int i = 0; i < NUM_ASTEROIDS; ++i)
    {
        Entity* entity = &game_state->asteroids[i];

        entity->size = asteroid_collision_radius;
        entity->rect.w = entity_size_world;
        entity->rect.h = entity_size_world;
        entity->texture_atlas = game_state->texture_atlas;
        entity->texture_rect.w = entity_size_texture;
        entity->texture_rect.h = entity_size_texture;
        entity->texture_rect.x = entity_size_texture * asteroid_sprite_x;
        entity->texture_rect.y = entity_size_texture * asteroid_sprite_y;
    }
}
```

```
// projectiles init
{
    for(int i = 0; i < NUM_PROJECTILES; ++i)
    {
        Entity* entity = &game_state->projectiles[i];

        entity->size = projectile_collision_radius;
        entity->rect.w = entity_size_world;
        entity->rect.h = entity_size_world;
        entity->texture_atlas = game_state->texture_atlas;
        entity->texture_rect.w = entity_size_texture;
        entity->texture_rect.h = entity_size_texture;
        entity->texture_rect.x = entity_size_texture * projectile_sprite_x;
        entity->texture_rect.y = entity_size_texture * projectile_sprite_y;
    }
}
```

literally the same code
different data

Exercise 01 review - Add projectiles (update)

```
// asteroids update

{ /* spawning logic */ }

for(int i = 0; i < NUM_ASTEROIDS; ++i)
{
    Entity* e = &game_state->asteroids[i];
    if(!e->alive)
        continue;

    e->position.y += context->delta * e->velocity;

    if(entity_collision_check(e, &game_state->player))
    {
        //SDL_Log("game over");
        game_state->game_over = true;
        return;
    }

    // despawn when out of bounds
    if(e->position.y > context->window_h + e->texture_rect.h)
        e->alive = false;
}
```

```
// projectiles update
for(int i = 0; i < NUM_PROJECTILES; ++i)
{
    Entity* e = &game_state->projectiles[i];
    if(!e->alive)
        continue;

    e->position.y -= context->delta * e->velocity;

    // despawn when out of bounds
    if(e->position.y < -e->texture_rect.h)
        e->alive = false;

    // despawn projectile and asteroids if they overlap
    for(int i = 0; i < NUM_ASTEROIDS; ++i)
    {
        Entity* asteroid_curr = &game_state->asteroids[i];
        if(!asteroid_curr->alive)
            continue;

        if(entity_collision_check(e, asteroid_curr))
        {
            e->alive = false;
            asteroid_curr->alive = false;
        }
    }
}
```

lots of similarities

Exercise 01 review - Beware shadowing!

ops, this is wrong!

but why does it work anyway here?

```
// projectiles update
for(int i = 0; i < NUM_PROJECTILES; ++i)
{
    Entity* e = &game_state->projectiles[i];
    if(!e->alive)
        continue;

    e->position.y -= context->delta * e->velocity;

    // despawn when out of bounds
    if(e->position.y < -e->texture_rect.h)
        e->alive = false;

    // despawn projectile and asteroids if they overlap
    for(int i = 0; i < NUM_ASTEROIDS; ++i)
    {
        Entity* asteroid_curr = &game_state->asteroids[i];
        if(!asteroid_curr->alive)
            continue;

        if(entity_collision_check(e, asteroid_curr))
        {
            e->alive = false;
            asteroid_curr->alive = false;
        }
    }
}
```

Exercise 01 review - Add projectiles (render)

```
// asteroids render
for(int i = 0; i < NUM_ASTEROIDS; ++i)
{
    Entity* asteroid_curr = &game_state->asteroids[i];
    if(!asteroid_curr->alive)
        continue;

    entity_render(context, asteroid_curr);
}
```

```
// projectiles render
for(int i = 0; i < NUM_PROJECTILES; ++i)
{
    Entity* entity = &game_state->projectiles[i];
    if(!entity->alive)
        continue;

    entity_render(context, entity);
}
```

literally the same code

Exercise 01 review - Spaw entities

```
// spawn a new entity
// NOTE: this works even if we are passing an array (ie, the original declaration of `GameState::asteroids` and `GameState::projectiles`
//       as `Entity asteroids[NUM_ASTEROIDS]), since they are *mostly* equivalent.
Entity* entity_spawn(Entity* pool, int max_amount, float x, float y, float velocity)
{
    // find id of first alive entity (checking we are not skipping past the end of the array
    int new_entity_idx = 0;
    while(new_entity_idx < max_amount && pool[new_entity_idx].alive)
        ++new_entity_idx;

    if(new_entity_idx == max_amount)
    {
        // NOTE: since we changed `GameState::asteroids` and `GameState::projectiles` from array of entities to pointers,
        //       in theory we could resize entity array here. It would require some additional changes to keep track of
        //       the new `max_amount` of entities in the pool (and we would need to set a ceiling anyway, we can't keep
        //       spawning new entities forever
        SDL_Log("[WARNING] too many entities, cannot spawn more!");
        return NULL;
    }

    Entity* entity = &pool[new_entity_idx];

    entity->alive = true;
    entity->position.x = x;
    entity->position.y = y;
    entity->velocity   = velocity;

    // return the newly spawned entity, so we can do additional initialization if needed
    return entity;
}
```

Exercise 01 review - Reset game state

can be done better

```
// beginning of init()

// reset game state
{
    // NOTE: from `SDL_Free` documentation: "if `mem` [the parameter] is NULL, this function does nothing
    //       ie, we don't need to test for null here (like most implementations of the standard `free`).
    SDL_free(game_state->asteroids);
    SDL_free(game_state->projectiles);

    // NOTE: from `SDL_DestroyTexture`: "Passing NULL or an otherwise invalid texture will set the SDL error message to "Invalid texture""
    //       ie, we COULD avoid this test, but if you're keeping track of SDL internal messages your logs will be full "errors" that will drown
    //       the real problems in a sea of noise, so let's avoid it
    if(game_state->texture_atlas)
        SDL_DestroyTexture(game_state->texture_atlas);
}
// zero out memory (need to do this explicitly if we want to reset the game state)
SDL_memset(game_state, 0, sizeof(GameState));

{ /* load textures */ }

// game state
{
    game_state->asteroids = (Entity*)SDL_calloc(NUM_ASTEROIDS, sizeof(Entity));
    game_state->projectiles = (Entity*)SDL_calloc(NUM_PROJECTILES, sizeof(Entity));
    game_state->asteroid_spawning_period = asteroid_spawning_period_base;
}

// entities below...
```

```
// beginning of main()
init(context, game_state);

// beginning of update()
if (game_state->game_over)
    init(context, game_state);
```

Exercise 01 review - Reset game state, improved

```
static void init(SDLContext* context, GameState* game_state)
{
    // zero out memory
    SDL_memset(game_state, 0, sizeof(GameState));

    { /* load textures */ }

    game_state->asteroids = (Entity*)SDL_calloc(NUM_ASTEROIDS, sizeof(Entity));
    game_state->projectiles = (Entity*)SDL_calloc(NUM_PROJECTILES, sizeof(Entity));

    // this has to be reset every time we reset the game
    // game_state->asteroid_spawning_period = asteroid_spawning_period_base;


    reset(context, game_state);
}
```

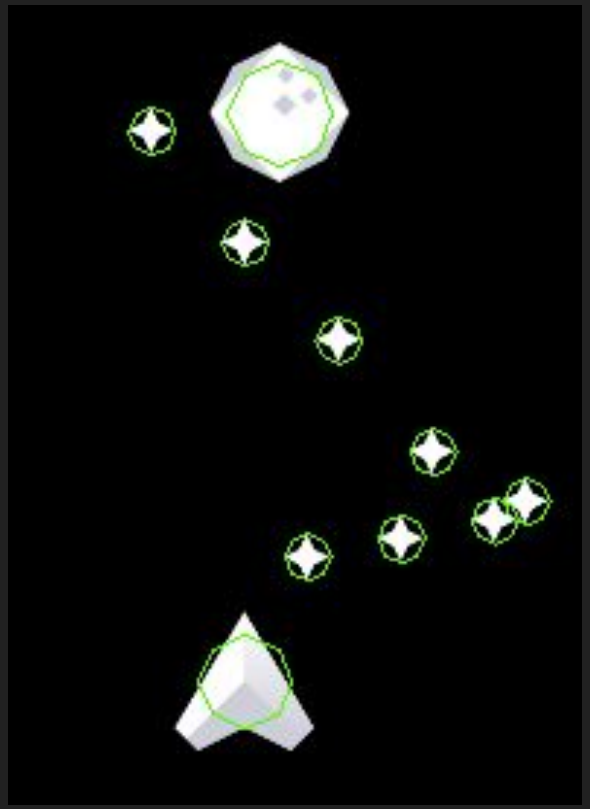
```
// beginning of main()
init(context, game_state);

// beginning of update()
if (game_state->game_over)
    reset(context, game_state);
```

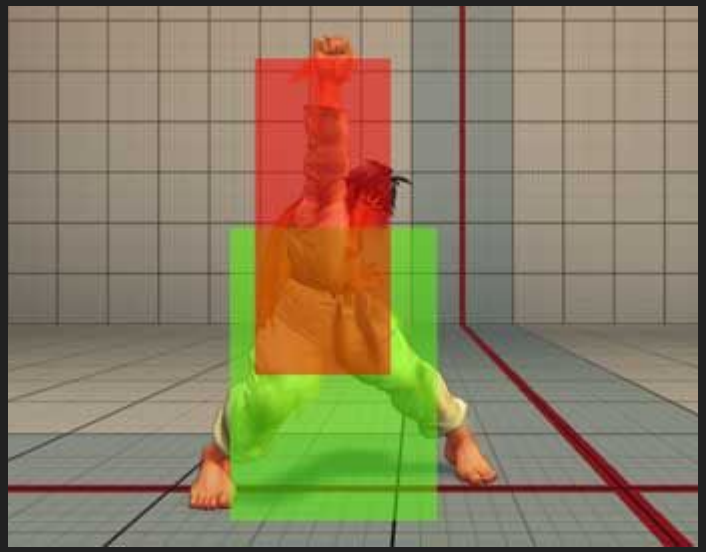
Exercise 01 review - handling projectile collisions

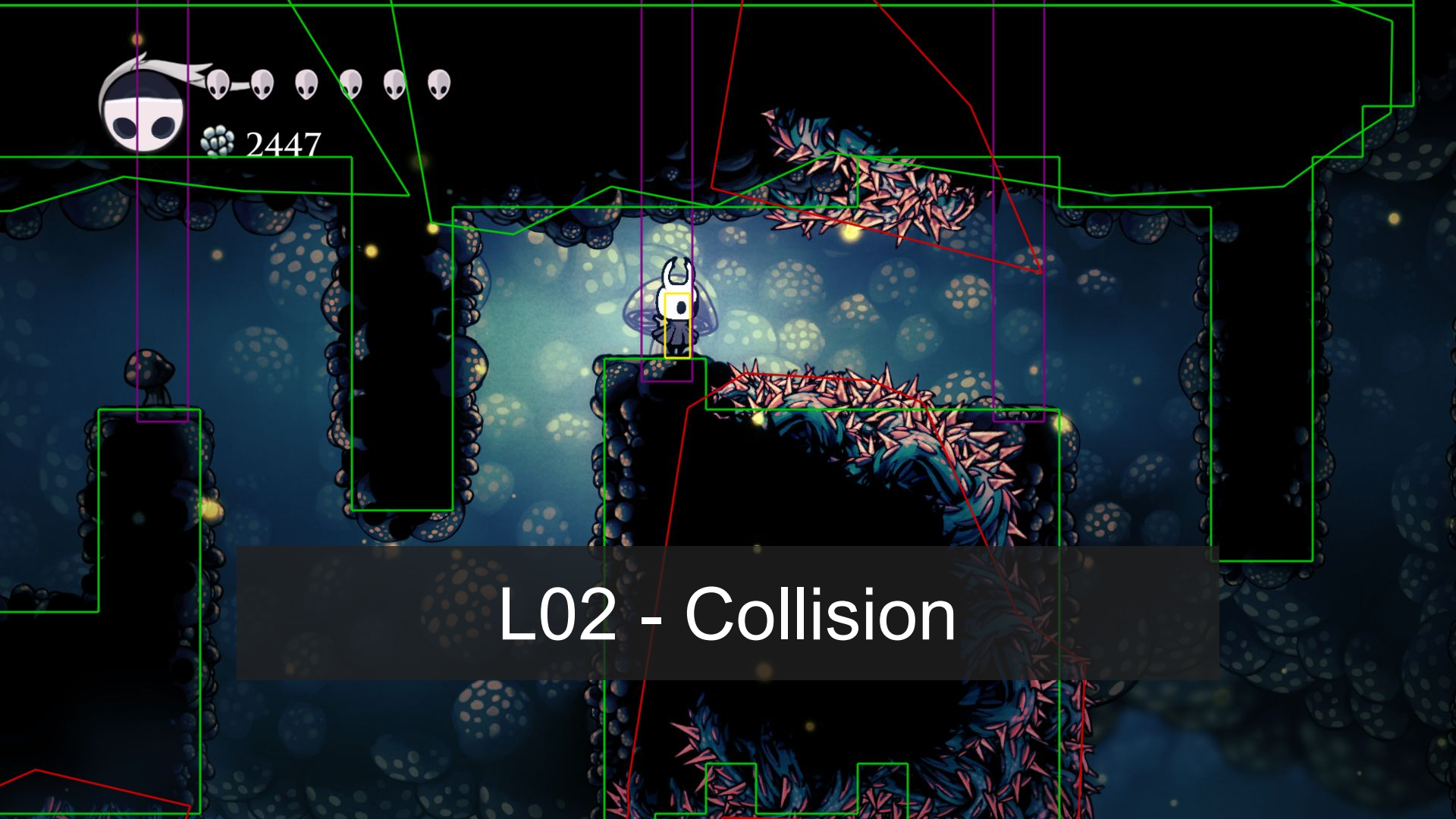
```
// projectile update
for(int i = 0; i < NUM_PROJECTILES; ++i)
{
    Entity* entity = &game_state->projectiles[i];
    if(!entity->alive)
        continue;

    // other update stuff...

    // actual collision detection
    for(int j = 0; j < NUM_ASTEROIDS; ++j)
    {
        Entity* asteroid_curr = &game_state->asteroids[j];
        if(!asteroid_curr->alive)
            continue;

        if(entity_collision_check(entity, asteroid_curr))
        {
            entity->alive = false;
            asteroid_curr->alive = false;
        }
    }
}
```

today’s main topic!

Exercise 01 review - collision size



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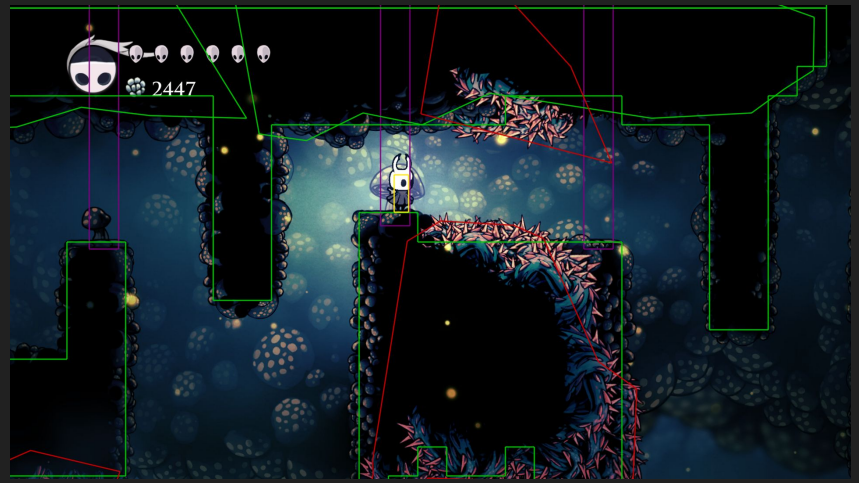
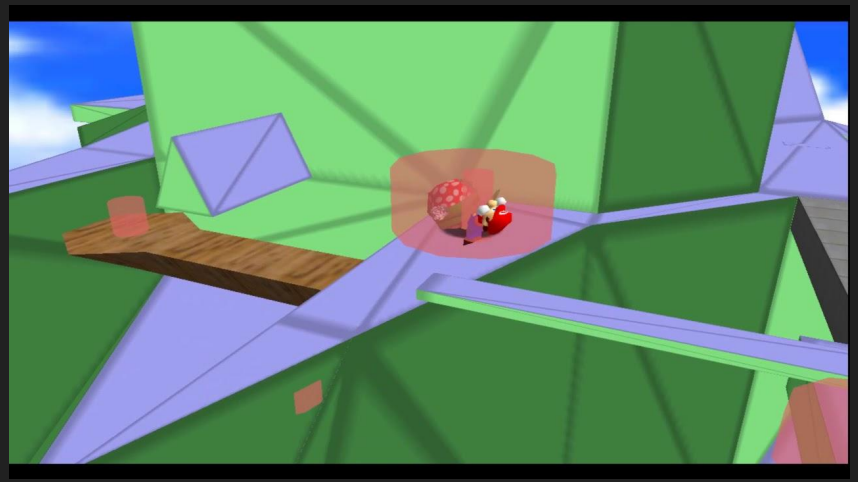
L02 - Collision

Problem statement: How do we…



- …efficiently detect collisions between arbitrary shapes?
- …tackle the quadratic nature of checking multiple colliders we saw in exercise 01?

- …separate entities that are colliding?



Agenda

1. Physics vs Collisions
2. Geometric overlaps and separations
3. Optimizing collision detection

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1. Physics vs Collisions
2. Geometric overlaps and separations
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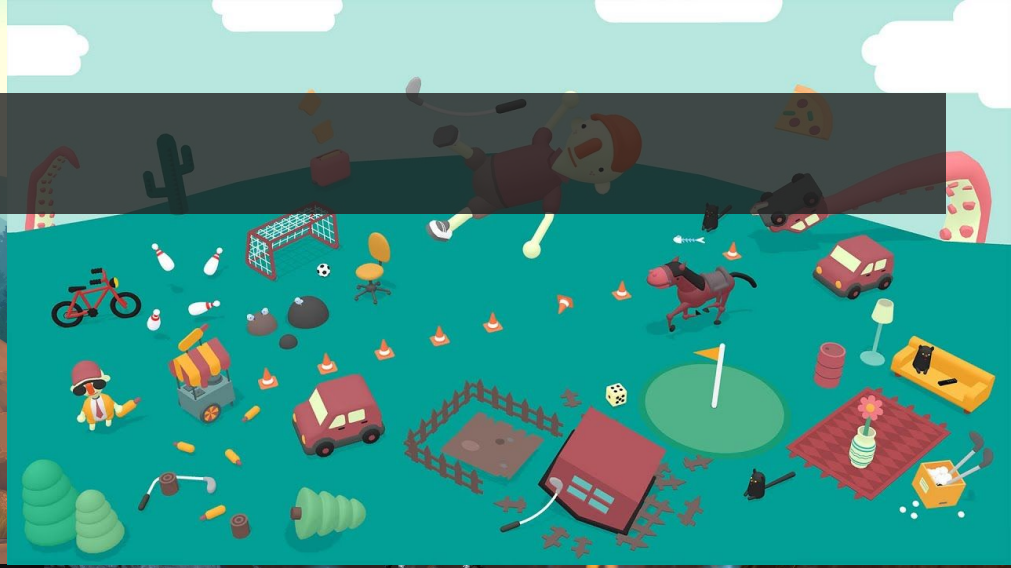
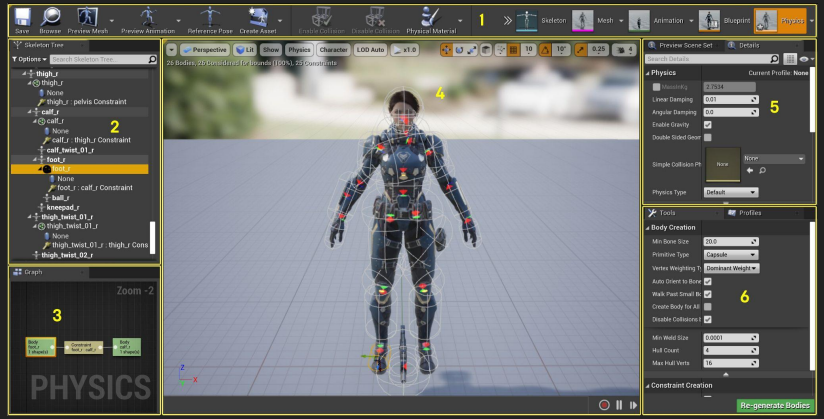
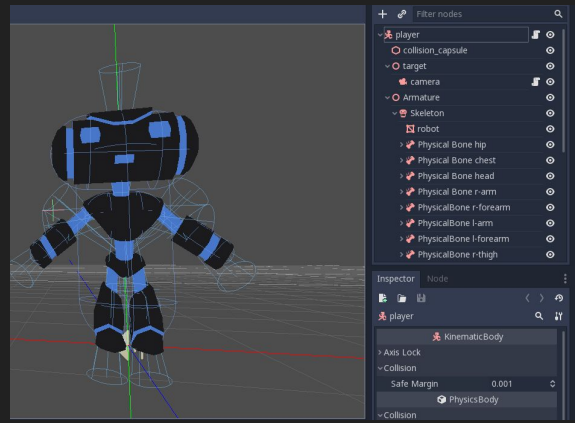
Simulation vs Control



Impact of physics on a Game

- Design
  - loss of control and predictability
  - emergent behaviours
- Engineering
  - more tools needed
  - all systems need to work with it accordingly
- Art
  - tools are more complex
  - content needs to be aware of physic
  - how does content interact with physics? (ie, animations vs physics)





3rd party Collision/Physics

- Open Dynamics Engine (ODE): Free and full source
- Bullet: Used for games and film, free and supports CCD (continuous collision detection)
- TrueAxis: free for non commercial use
- Jolt Physics: MIT licence, a few AAA games shipped with it
- PhysX: nVIDIA’s free lib (for GPU co-processing), open source since 2018
- Havok: Commercial (3D) gold standard, pay to use
- Box2D: 2D physics and collision library, free, open source

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- Box2D: 2D physics and collision library, free, open source

Good options for your final project if you want 3D

used in L04 Physics

Collision detection vs physics simulation.

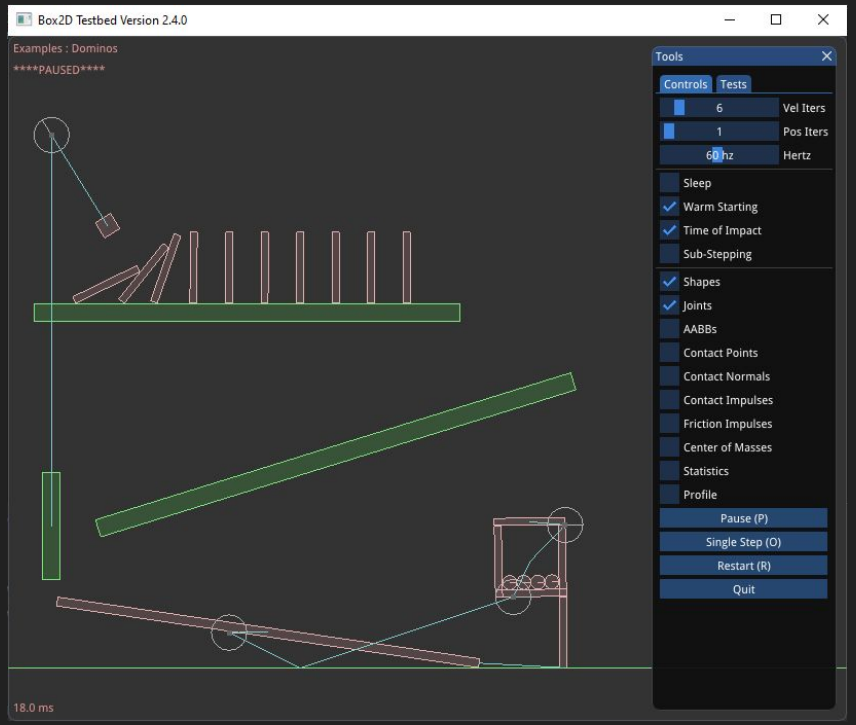
- Physics simulations depends strongly on efficient collision detection.
- APIs often have clean separation between the two concepts.

Agenda

1. Physics vs Collisions
2. Geometric overlaps and separations
3. Optimizing collision detection

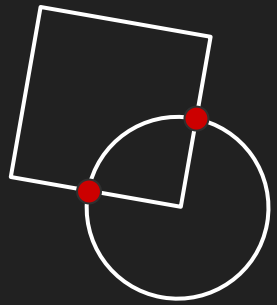
The Glorified Geometric Intersection Tester?

- The collision system detects if any pairs objects collide
- Objects are represented by simplified primitive shapes
- If a collision is detected contact information is generated, to allow object separation

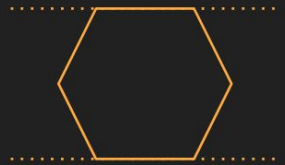


Collision Terms

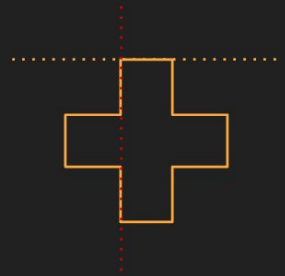
- Contact (or collision)
- Intersection (or overlap/penetration)
- Convexity (Convex vs Concave)



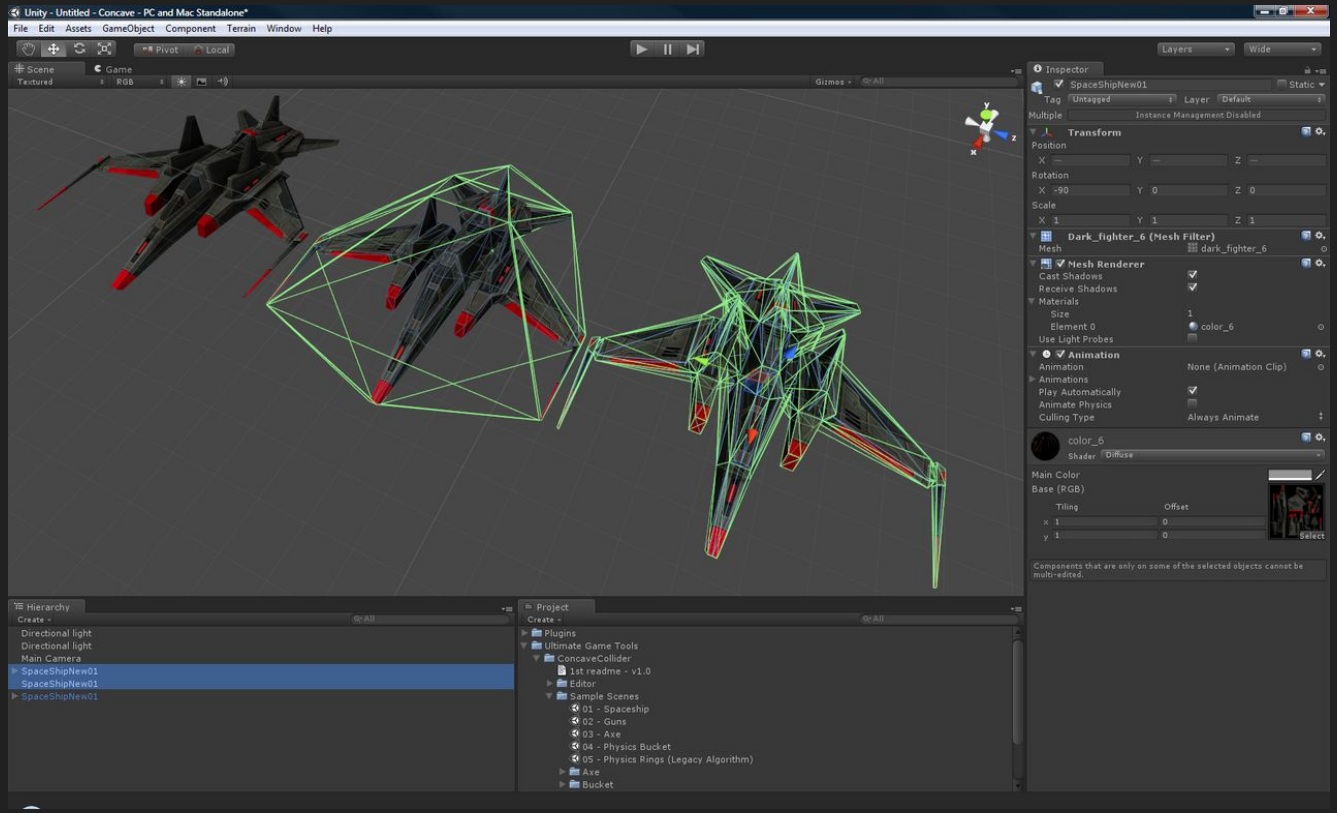
Convex polygon



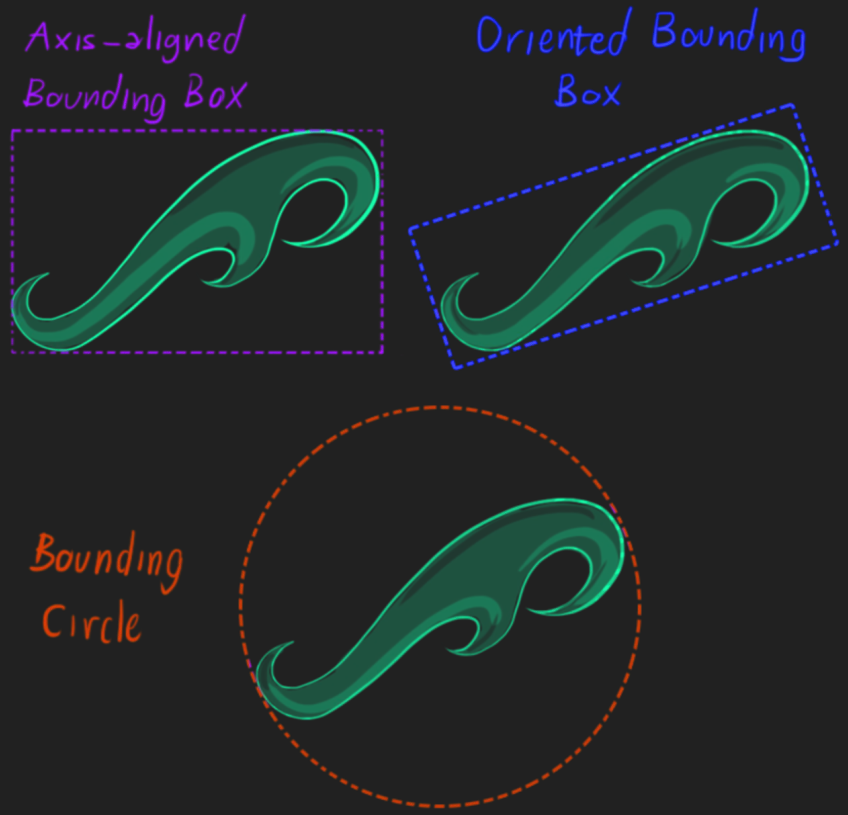
Concave polygon



Collision representations

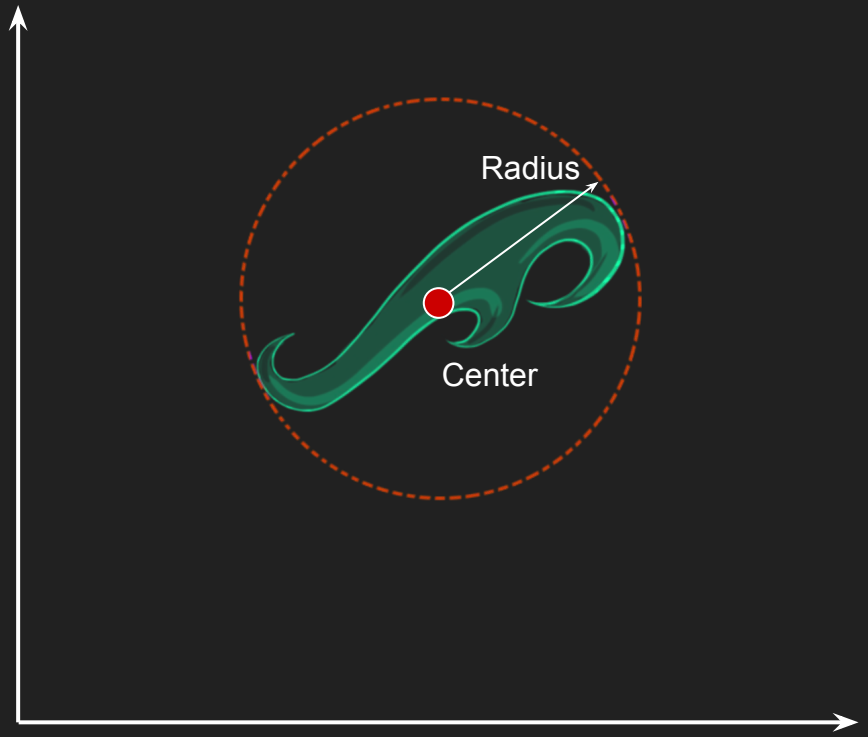


Bounding representation



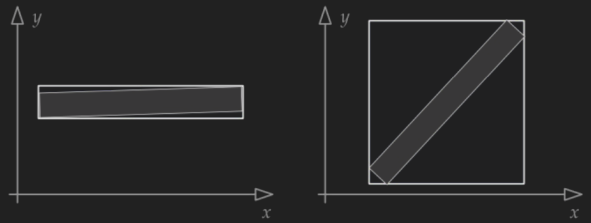
Bounding circle

- Represented by
  - Center position
  - Radius
- A point collides with bounding circle if distance to center is less than radius: `|center-p| < radius`



Axis-Aligned Bounding Box (AABB)

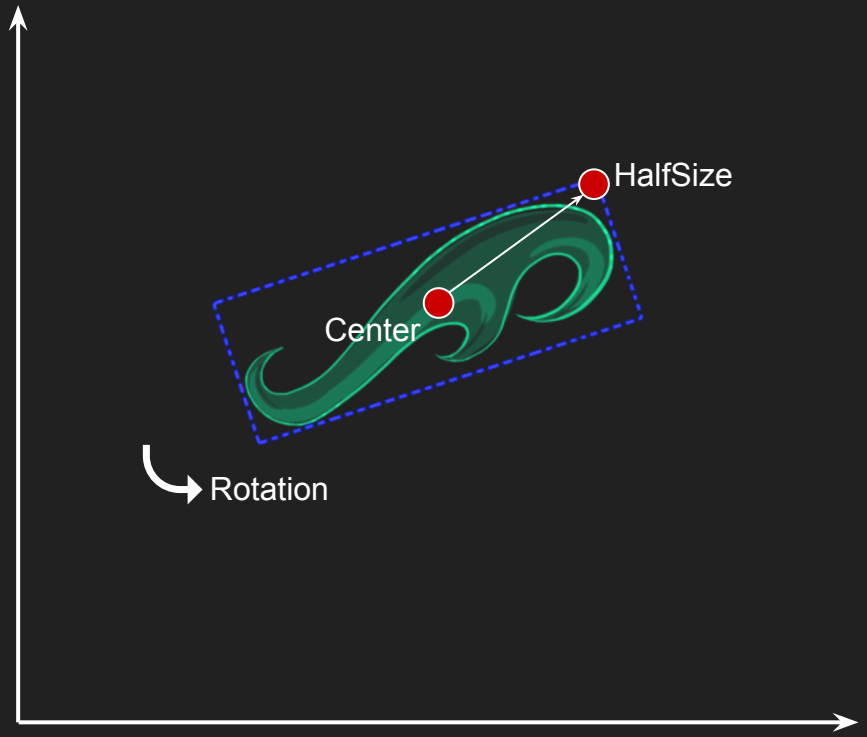
- Represented by either
  - min + max positions
  - Center position + half-size
- Has to redefined if object rotates



- A point collides with AABB if larger than min and less than max:
  min < p < max

Oriented Bounding Box (OBB)

- Represented by
  - Center position
  - half-size
  - rotation



Collision Primitives

Fastest

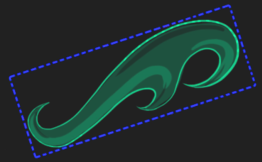
1. Spheres (or Circles in 2D)
2. Capsules
3. Axis-Aligned Bounding Boxes (AABB)
4. Oriented Bounding Boxes (OBB)
5. Discrete Oriented Polytopes (DOP)
6. Arbitrary Convex Volumes
7. Poly Soup

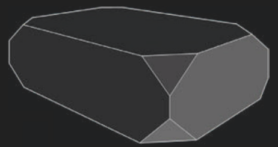
Slowest

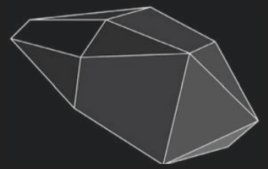
1 

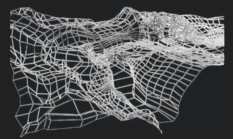
2 

3 

4 

5 

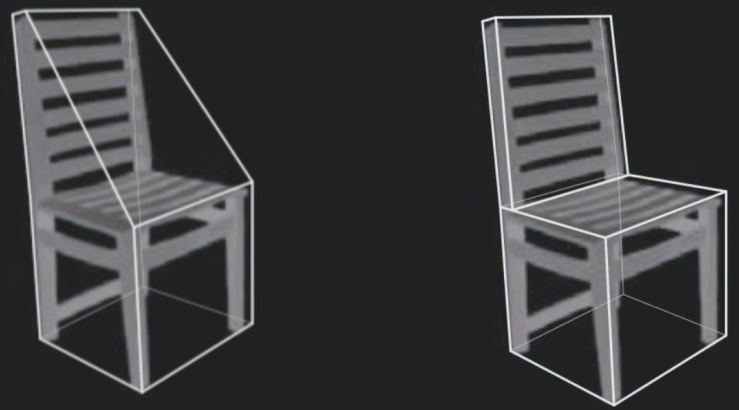
6 

7 

Compound shapes

- A concave object can be approximated by multiple convex colliders

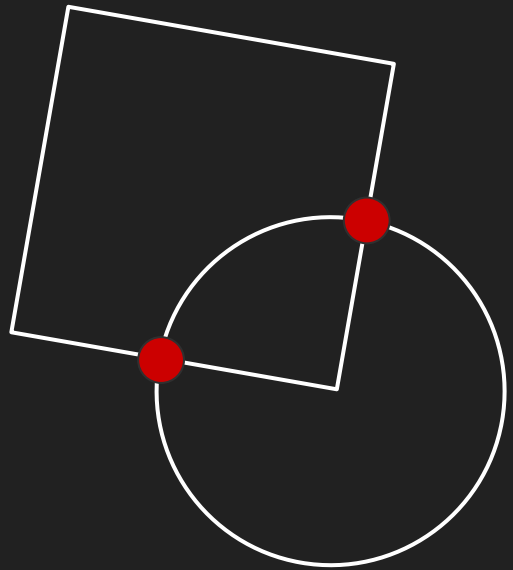
- Convex Decomposition: The process of cutting a concave object into convex parts



Collision testing

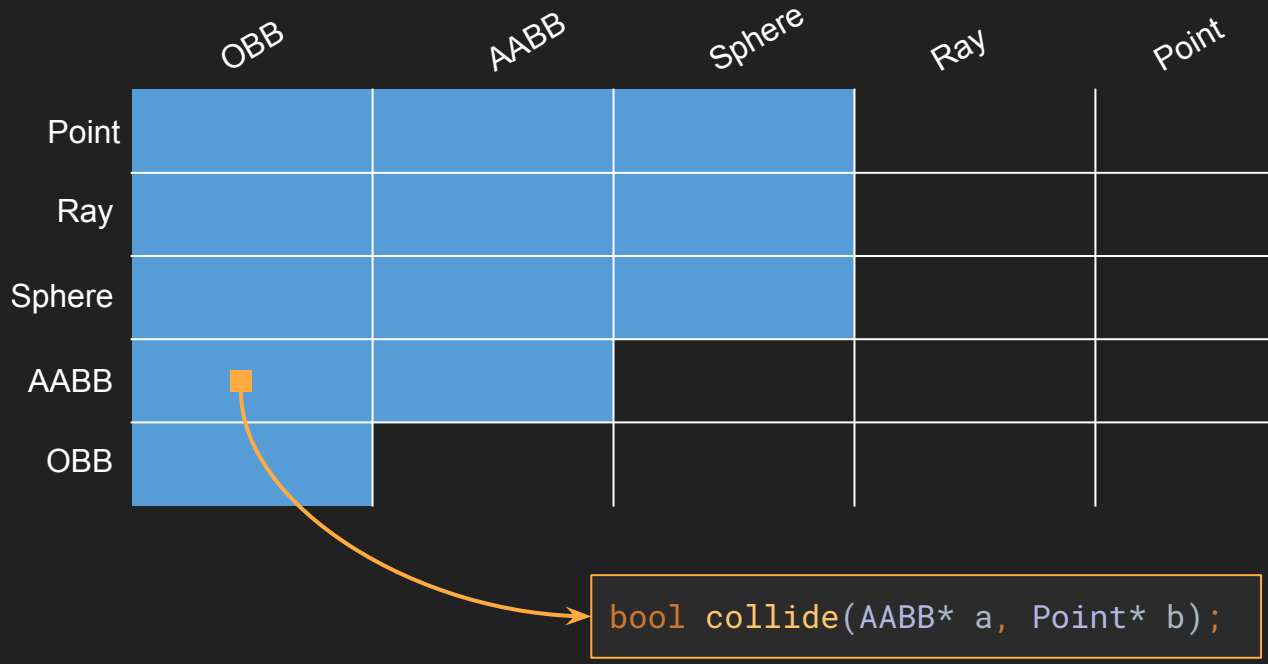
Determine if two collision primitives collide (and how)

- Point vs Sphere
- Sphere vs Sphere
- AABB vs Point
- AABB vs Sphere
- AABB vs AABB
- Etc.

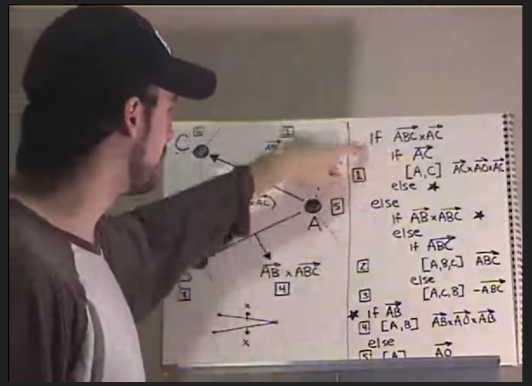


Collision testing

Physics system need to be able to test against pairs of collision primitives



GJK Algorithm



Simple but powerful algorithm to handle arbitrary convex shape.

References:

- Original paper: https://graphics.stanford.edu/courses/cs448b-00-winter/papers/gilbert.pdf
- Good intuitive explanation: https://caseymuratori.com/blog_0003
- Modern stuff (also hardcore simulations, not just games)
  - https://arxiv.org/pdf/2205.09663
  - https://graphics.stanford.edu/courses/cs468-01-fall/Papers/van-den-bergen.pdf
  - https://ora.ox.ac.uk/objects/uuid:69c743d9-73de-4aff-8e6f-b4dd7c010907

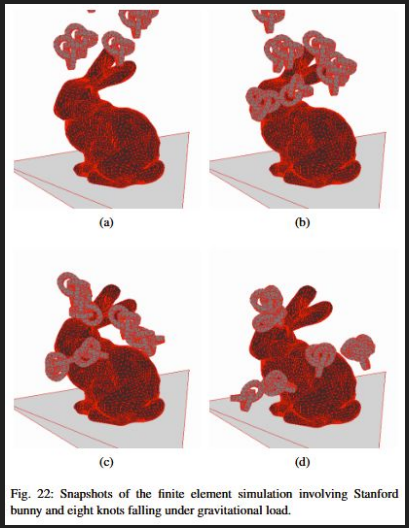
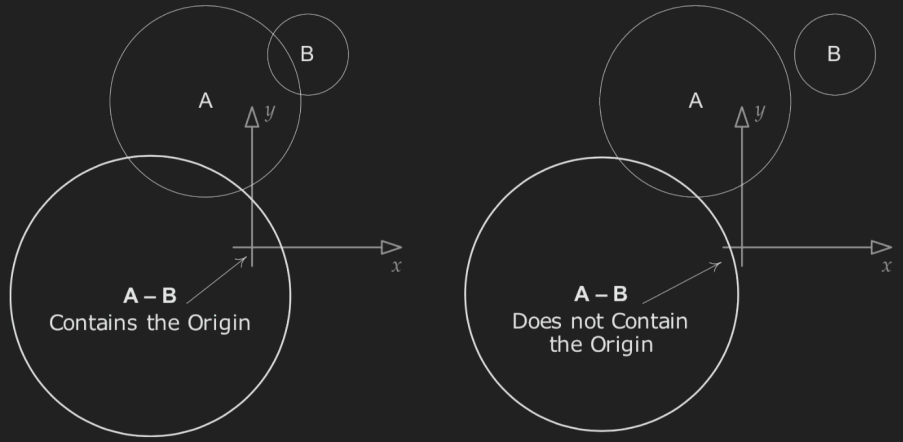


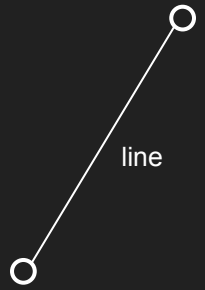
Fig. 22: Snapshots of the finite element simulation involving Stanford bunny and eight knots falling under gravitational load.

GJK Insights - Minkowsky difference

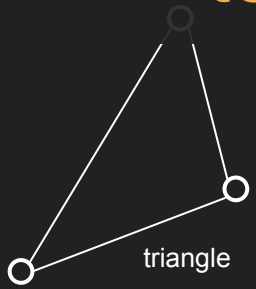
- geometrical entity constructed by taking the difference between "all points" two different shapes
- The Minkowsky difference contains the origin IIF the two original shapes are colliding
- Two main questions tho:
  - how do we do this operation efficiently?
  - how do we test if an arbitrary (convex) shape contains the origin?

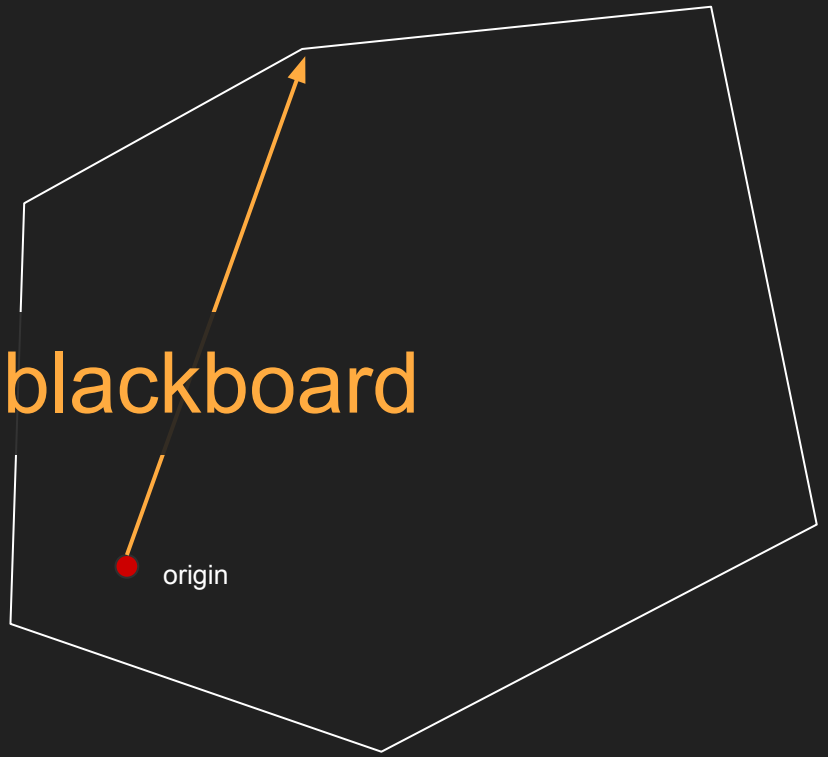


GJK Insights - support function and simplex



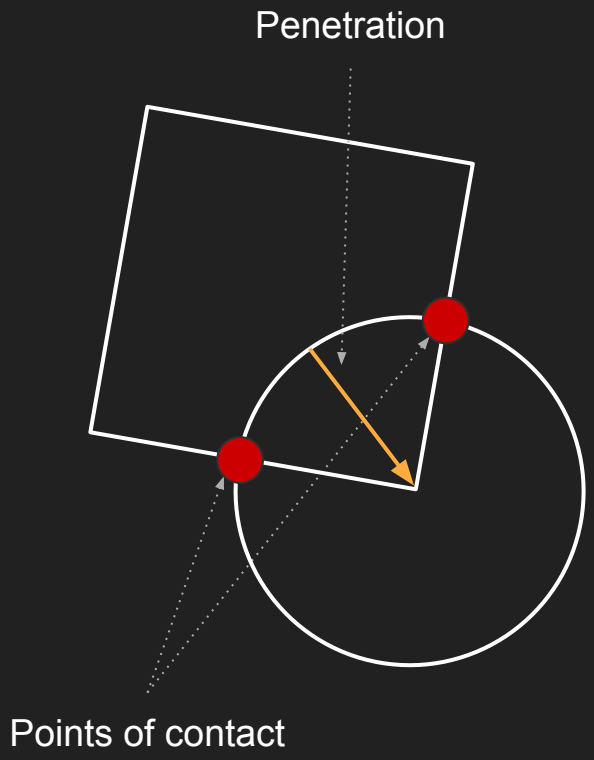
to the (virtual) blackboard





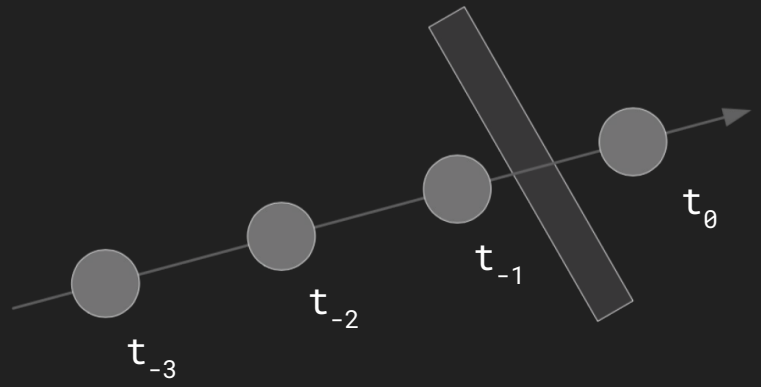
Separation

- Calculate the point of contact
- Calculate collision direction
- Calculate overlapping depth
- Use that for separation, eg. move one object back



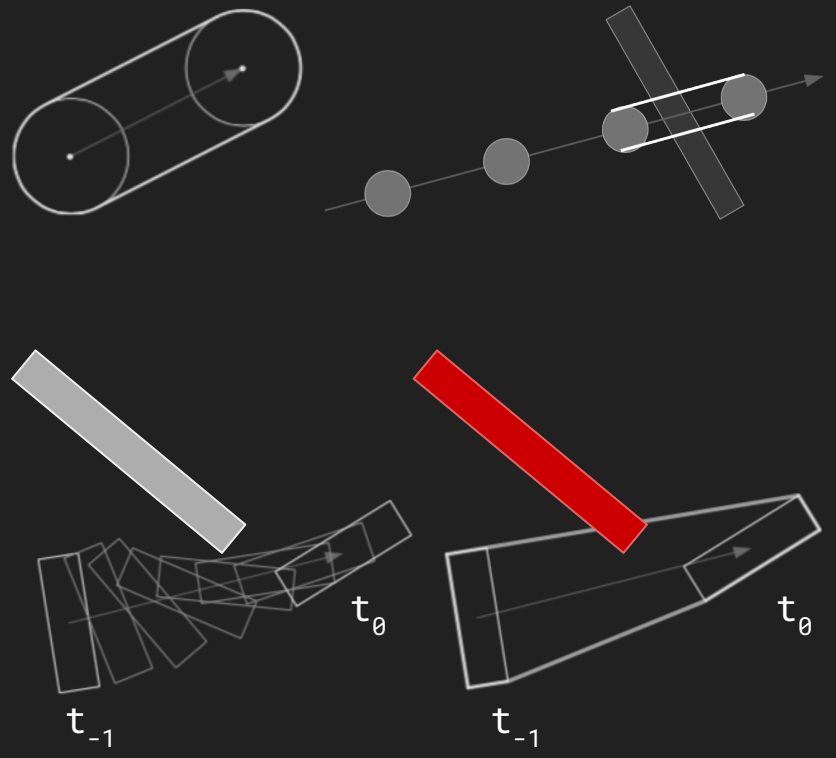
Discrete detection

- At each simulated timestep, move all objects to their new locations, then do static testing
- Works well for objects that move slowly relative to their size, and is default in most engines
- Fast movement relative to size produce gaps, which means small objects than be “overlooked”



Swept Shapes

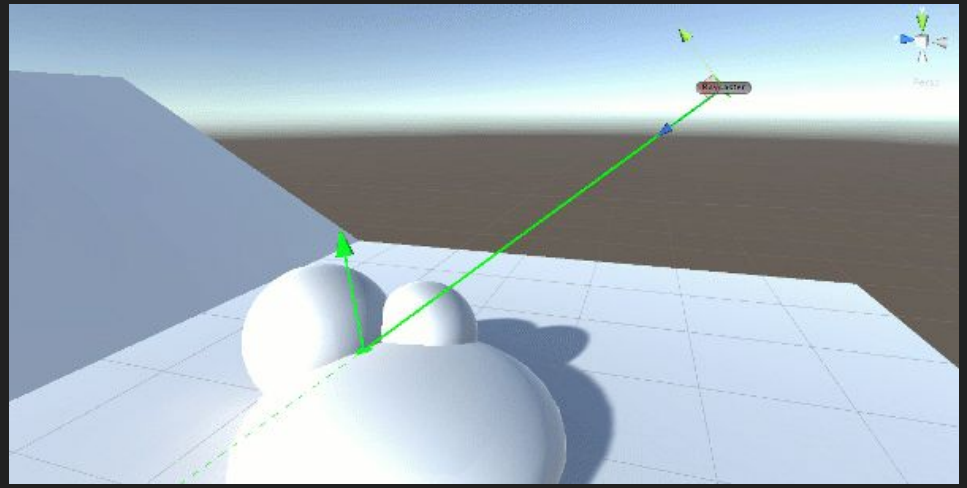
- Swept Sphere becomes a Capsule
- Then calculate static collisions on the resulting shapes
- Issues with objects that follow curved path or that are rotating, might create a concave shape.

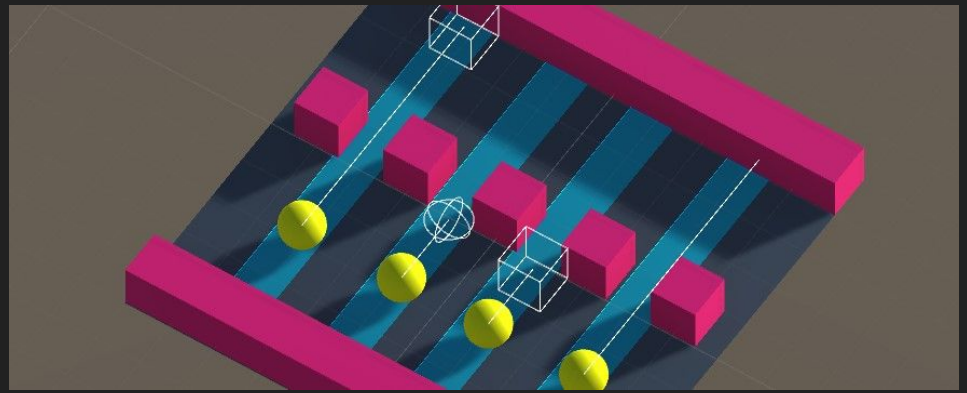


Collision Queries

Hypothetical collision tests (no impact, just info)

- Point test
- Ray casting
- Shape casting





Agenda

1. Physics vs Collisions
2. Geometric overlaps and separations
3. Optimizing collision detection

Optimizing collision detection

Finding collisions between all objects is expensive

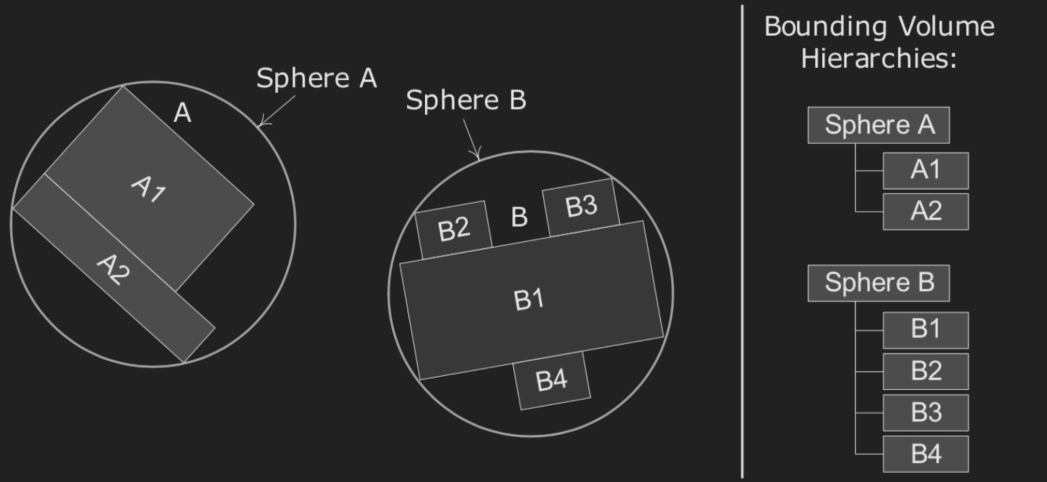
- because a single call to check collision can be expensive (see previous slides)
- because we are doing a lot of them (naive approach: $O(n^2)$)

```
for(int i = 0; i < entities_alive_count; ++i)
    for(int j = 0; j < entities_alive_count; ++j)
        if(entity_collision_check(entities[i], entities[j]))
        {
            // handle collision
        }
```

Collision detection optimizations

- Pairs optimization
  - Bounding volumes hierarchies
- Global optimization
  - Static colliders
  - Collision layers
  - World partitioning

Is it worth running the full collision test between A and B?



test complex shapes only if simple bounding sphere/boxes are colliding

Collision detection optimizations

- Pairs optimization
  - Bounding volumes hierarchies
- Global optimization
  - Static colliders
  - Collision layers
  - World partitioning

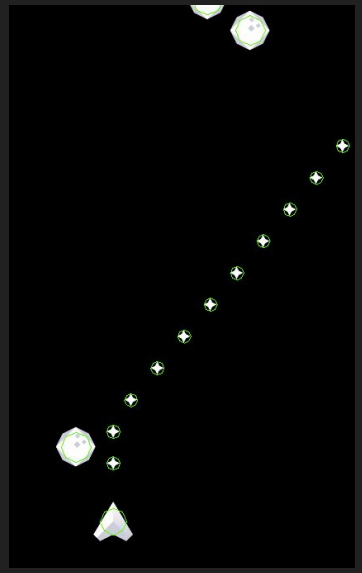
Immobile objects like platforms, ground etc will never collide, unless someone collides with them

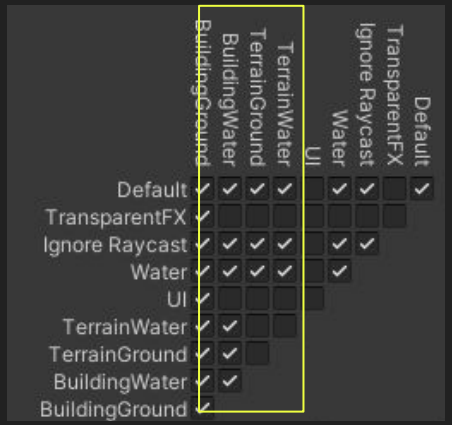
test dynamic colliders VS all colliders (instead of all VS all)

Collision detection optimizations

- Pairs optimization
  - Bounding volumes hierarchies
- Global optimization
  - Static colliders
  - Collision layers
  - World partitioning

Do we care about collisions between A and B?





Collision detection optimizations

- Pairs optimizati
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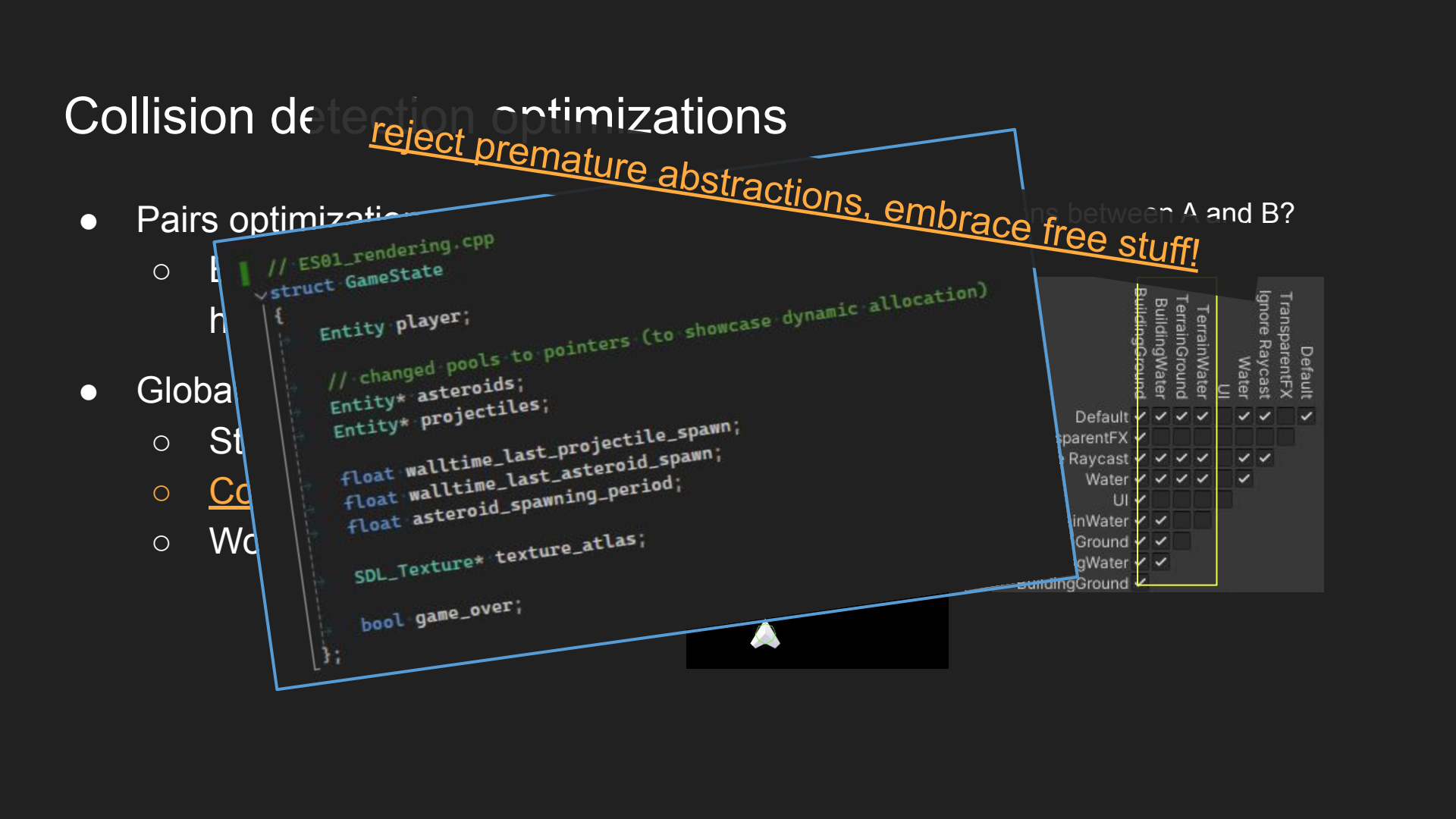
- Globa
  - St
  - Co
  - Wo

reject premature abstractions, embrace free stuff!

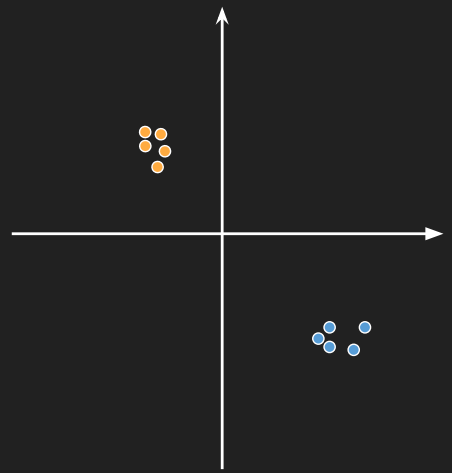
```cpp
// ES01_rendering.cpp
struct GameState
{
    Entity player;

    // changed pools to pointers (to showcase dynamic allocation)
    Entity* asteroids;
    Entity* projectiles;

    float walltime_last_projectile_spawn;
    float walltime_last_asteroid_spawn;
    float asteroid_spawning_period;

    SDL_Texture* texture_atlas;

    bool game_over;
};
```



Collision detection optimizations

- Pairs optimization
  - Bounding volumes hierarchies
- Global optimization
  - Static colliders
  - Collision layers
  - World partitioning

Is there ANY chance these objects can collide?

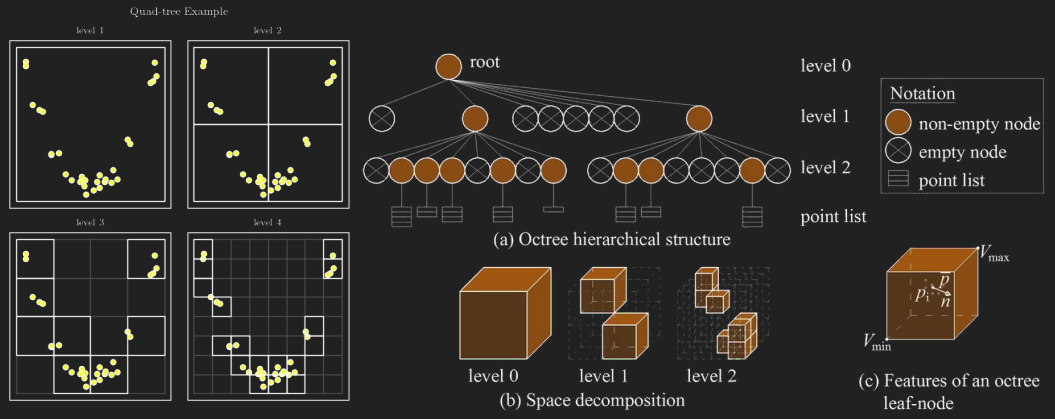


spit the world in different regions, test only objects in the same region

World Partitioning algorithms

- Quadtrees/Octrees
- Dynamic AABB Trees
- Others

- Recursively subdivide space in equal size regions.
- Regions will be lower resolutions for regions with less objects
- Perform collision test only between objects in the same region
- Remember, objects can belong to multiple regions when they cross the boundaries

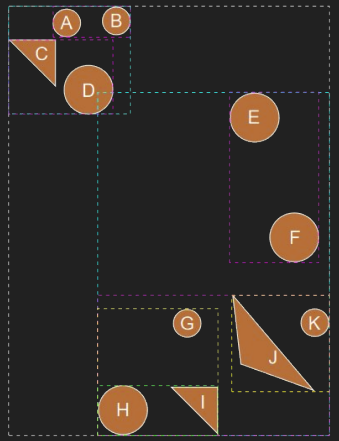


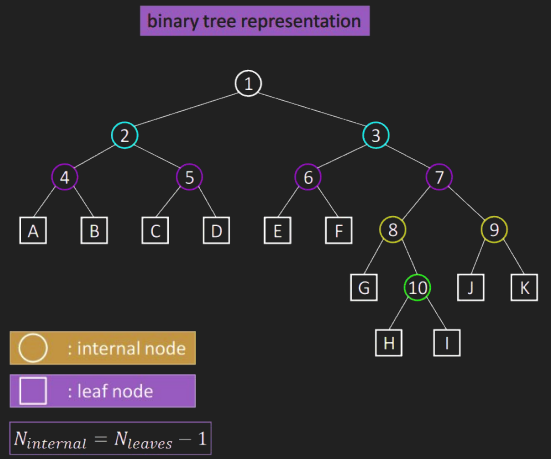
Interactive explanation: https://jimkang.com/quadtreevis/

World Partitioning algorithms

- Quadtrees/Octrees
- Dynamic AABB Trees
- Others

- Recursively group objects' AABB into pairs
- Leaves are objects' AABB, internal nodes are union of children's AABB
- If object is not colliding with a (internal) node, it can't collide with any children!
- can be extremely effective, but keeping the tree balanced is key

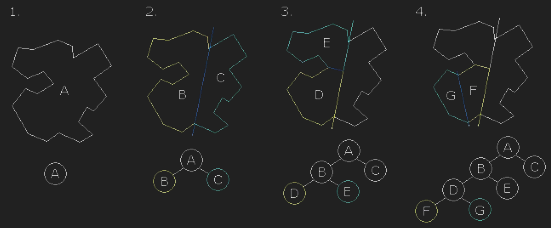




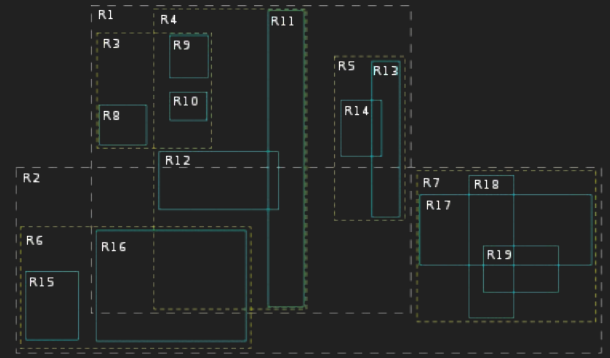
https://box2d.org/files/ErinCatto_DynamicBVH_GDC2019.pdf

World Partitioning algorithms

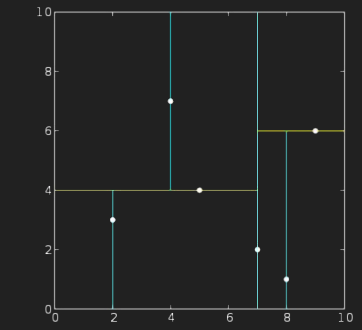
- Quadtrees/Octrees
- Dynamic AABB Trees
- Others



Binary Partition Tree (BPS)



R-Tree



k-d Tree

https://en.wikipedia.org/wiki/Space_partitioning#In_computer_graphics

Collision detection revised

Collision detection is now performed in two steps:

- Broad-phase: separate objects into multiple groups
- Narrow-phase: intersection tests on all objects within this groups

To study for the exam

- types of colliders
- generic overview of the GJK algorithm
- difference between broad and narrow phase
- ways to optimize collision detection
- overview of world partition algorithms

Today’s exercise